STEMMate A4 Testing Report

SDN621S Software Design - Role 7 Tester

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1. Summary

Forty automated Java tests were executed against the supplied repository. Thirty passed and ten failed. The tests include real file storage and separate-process restart checks. Failures are kept in the evidence and defect log. The result does not establish that the complete application is ready for acceptance.

Test group	Executed	Passed	Failed
Desktop client	25	18	7
Java server	11	9	2
Standalone prototype	4	3	1
Total	40	30	10

2. Scope and method

The runner compiles 45 selected production-core and test source files in OpenJDK 21.0.13. It runs every case in a separate JVM with an isolated data folder. It uses the real services, repositories, models, file stores, validation and route-handler code. Remote-store and HTTP exchange/client doubles are identified in the relevant cases. All data is synthetic.

The suite uses a small Java assertion runner with repeatable pass/fail output. JUnit-format XML is provided as an evidence format; the JUnit library is not used. No Maven download is required for this runner. The complete Swing GUI, SQLite JDBC runtime and live client/server network were not executed.

3. Main findings

Normal restart persistence passed for the desktop plan, draft, outbox and saved-offline mark, and for the standalone prototype text outbox.

The desktop JSON request contains unquoted string values. Its acknowledgement parser can return a rejected ID as confirmed.

Failed writes and interrupted SYNCING records have recovery gaps. Required safety content and Java-server FR4 checks are incomplete.

The desktop/Ballerina JSON path and standalone-prototype/Java-server line path are different. The final matching pair must be stated in the handover.

Full case steps, expected results, observed output and requirement/component mappings are in docs/tests/role7/TEST_CASES.md and TRACEABILITY_MATRIX.md. The report uses the existing FR, QR, SUS and AC identifiers from the repository.

4. Desktop client tests J01 to J13

Each result refers to the stated assertion. Evidence is in docs/tests/role7/evidence/client, with one log per test and an aggregate results.json.

Test	Requirement	Check	Result	Issue
J01	FR1	Combined catalogue filters	PASS	-
J02	FR2	Offline activity selection persists	PASS	-
J03	FR3, FR4	Draft is created from the selected activity	PASS	-
J04	FR3, FR4, FR5	A valid plan is saved and queued	PASS	-
J05	FR3, FR4	Blank title is rejected	PASS	-
J06	FR4	Missing required plan elements are rejected	PASS	-
J07	FR4	Negative and mismatched timing are rejected	PASS	-
J08	FR6	Incomplete draft can be autosaved and restored	PASS	-
J09	FR2, FR3, FR5, FR6, QR1	Plan draft and outbox survive a real JVM restart	PASS	-
J10	FR3, FR5	Saving the same plan ID is idempotent locally	PASS	-
J11	FR5, QR1	Offline state prevents remote calls	PASS	-
J12	FR5	Reconnect synchronises a queued plan	PASS	-
J13	FR5, QR1, SUS1	Failure retains data logs the error and permits retry	PASS	-

J09 is an actual normal-process restart test. The writer process stores the complete plan, draft, offline mark and queued record, then exits. A different reader process verifies the stored values using the same isolated user.home. This checks the application file stores rather than a mocked storage Map.

J11-J15 use a remote-store double to control offline, reconnect, partial acknowledgement, failure and batch behaviour. They do not prove a live server upload.

5. Desktop client tests J14 to J25

Each result refers to the stated assertion. Evidence is in docs/tests/role7/evidence/client, with one log per test and an aggregate results.json.

Test	Requirement	Check	Result	Issue
J14	FR5, QR1	Only acknowledged plans are confirmed	PASS	-
J15	QR2, FR5	Sync batches are bounded at twenty	PASS	-
J16	FR4, FR5	HTTP payload quotes scalar JSON strings	FAIL	D01
J17	FR5	Valid acknowledgement IDs are parsed	PASS	-
J18	FR5, QR1	Rejected acknowledgement IDs must not be accepted	FAIL	D02
J19	FR5, QR1	HTTP failure becomes a retryable network error	PASS	-
J20	QR2	Oversized HTTP batch is rejected before send	PASS	-
J21	FR3, QR1	Failed plan write must not leave an orphan outbox	FAIL	D03
J22	FR6, QR1	Wrong stored object type is handled on load	FAIL	D04
J23	FR4	Whitespace safety content is rejected	FAIL	D05
J24	FR5, QR1	Interrupted SYNCING record remains retryable after reload	FAIL	D06
J25	AC1	Swing form labels have explicit label bindings	FAIL	D07

J16-J20 use a capturing HttpClient double while executing the actual HTTP adapter. The request body and response parsing are inspected without opening a socket. J21 and J22 inject failures using real files. J24 checks a real persisted interrupted state. J25 is a source audit for explicit label associations, not a screen-reader test.

6. Server and standalone prototype tests

Server cases execute validation, encoding and route dispatch. HttpExchange is a test double; no live HTTP server or SQLite database is opened. Prototype cases execute ServerClient codecs and the real text-file Outbox.

Test	Requirement	Check	Result	Issue
S01	FR3, FR4	Server accepts a valid synthetic plan	PASS	-
S02	FR3, FR4	Server rejects blank identity title and safety	PASS	-
S03	FR3	Server enforces the 120 character title limit	PASS	-
S04	FR5	Server rejects invalid update timestamps	PASS	-
S05	FR3	Server rejects a malformed activity ID	PASS	-
S06	FR4	Server checks every FR4 element	FAIL	D08
S07	FR3, FR5	Server line codec preserves special characters	PASS	-
S08	SUS1	Health route returns its documented status	PASS	-
S09	FR5	Sync route rejects invalid device ID	PASS	-
S10	QR2	Sync route rejects an oversized body	PASS	-
S11	FR5	Java server supports the desktop upload route	FAIL	D09
P01	FR3, FR5	Prototype line codec preserves plan text	PASS	-
P02	FR5, QR1	Prototype outbox writes and reloads exact lines	PASS	-
P03	FR5, QR1	Prototype outbox survives a real JVM restart	PASS	-
P04	FR5, QR1	Outbox write failure reaches its caller	FAIL	D10

S11 is a cross-component compatibility check. It shows that the main desktop client cannot simply be pointed at the supplied Java server without adaptation. The standalone prototype uses the Java server protocol, so this failure must not be described as evidence that the correctly paired prototype cannot sync.

P03 is a second actual separate-process restart test. It verifies the queued text line after the writer JVM has exited. P04 records a write failure that is printed to stderr but not returned to the calling code.

7. Priority defect records

D01 Desktop upload body is invalid JSON

Test: J16. Requirement: FR4, FR5. Severity: High. Owner: Role 5 / Role 6.

Actual: UUIDs, titles and timestamps are appended without JSON quotes. The captured request body contains title:Synthetic water plan without a string value.

Recommended action: Separate string-field serialization from raw array/number fields. Use a JSON writer or quote every scalar string. Retest J16 against the intended Ballerina API.

D02 Rejected IDs can be parsed as accepted

Test: J18. Requirement: FR5, QR1. Severity: High. Owner: Role 5 / Role 6.

Actual: A body with accepted=[] and the plan ID only under rejected returns that ID as confirmed.

Recommended action: Parse the accepted array structurally and accept only sent IDs inside that array. Reject malformed or inconsistent confirmations.

D03 Failed plan save leaves a durable orphan queue entry

Test: J21. Requirement: FR3, QR1. Severity: High. Owner: Role 4 / Role 5.

Actual: When plans.ser is made unwritable as a directory, save throws, but outbox.ser already contains one pending record without a durable plan.

Recommended action: Make plan and outbox persistence one durable transaction or use a recoverable journal. Do not describe synchronized methods as a cross-file transaction.

D06 Interrupted SYNCING records are excluded from retry

Test: J24. Requirement: FR5, QR1. Severity: High. Owner: Role 4 / Role 5.

Actual: A persisted SYNCING record reloads with retry queue size zero, because only WAITING_TO_SYNC and FAILED_RETRY are selected.

Recommended action: On startup recover interrupted SYNCING items to a retryable state, or explicitly include them in a safe idempotent retry strategy.

D09 Desktop client and Java server use incompatible upload contracts

Test: S11. Requirement: FR5. Severity: High if paired. Owner: Role 1 / Role 5 / Role 6.

Actual: The desktop client targets the Ballerina JSON POST /plans interface on port 9090. The supplied Java server accepts line-format POST /sync on port 8080 and returns 404 for POST /plans. The separate prototype client uses /sync.

Recommended action: The PM and developers must name the final client/server pair. If the desktop client is paired with the Java server, adapt the route, payload, IDs and timestamps. Do not treat this failure as proof that the correctly paired standalone prototype cannot sync.

8. Further defect records

D04 Unexpected stored object type crashes startup

Test: J22. Requirement: FR6, QR1. Severity: High. Owner: Role 5 / Role 6.

Actual: Loading a serialized String instead of a LinkedHashMap throws ClassCastException.

Recommended action: Check the restored type, preserve the original file, and return a clear recovery result or visible warning. Do not silently overwrite the old data.

D05 Blank safety-note content passes validation

Test: J23. Requirement: FR4. Severity: Medium. Owner: Role 5.

Actual: A nonempty list containing SafetyNote(" ") passes validation.

Recommended action: Validate the text of each required element as well as collection size. Apply equivalent checks to steps, materials and inclusion.

D07 Planner label bindings are missing in source

Test: J25. Requirement: AC1. Severity: Medium. Owner: Role 6.

Actual: The inspected form source contains zero JLabel.setLabelFor calls. Some sync/list controls have accessible names; that does not associate planner captions with their fields.

Recommended action: Bind labels to their actual fields and provide accessible names where needed. Then run keyboard and screen-reader checks. This static result is not a full accessibility verdict.

D08 Java server accepts plans missing FR4 content

Test: S06. Requirement: FR4. Severity: High. Owner: Role 4 / Role 5.

Actual: Removing steps, timing, materials and inclusion produces an empty validation-problem list. The current plan table also has no steps, timing or materials columns.

Recommended action: Agree the complete FR4 payload for the selected final slice, validate it at the server, and preserve every required element in durable storage.

D10 Prototype outbox does not report write failure to its caller

Test: P04. Requirement: FR5, QR1. Severity: High. Owner: Role 5 / Role 6.

Actual: An injected filesystem failure is printed to stderr, but save returns normally and gives the calling UI no exception or result indicating failure.

Recommended action: Return a success/failure result or throw a handled storage exception. Only show a durable-save message after a successful write.

All ten records are OPEN in the supplied production source. Recommended fixes have not been implemented or retested in this handover. The detailed defect log contains reproduction steps and links to the individual evidence logs.

9. Restart accessibility and remaining checks

Verified restart evidence

J09: one JVM wrote the complete plan, draft, pending outbox record and saved-offline activity mark. After it exited, a different JVM read and verified them. P03 performed the same write-exit-read sequence for the standalone prototype text outbox. Both passed.

This establishes normal application file persistence. It does not establish SQLite durability, a real power-cut recovery guarantee, cross-file atomicity or live server acknowledgement. J21 and J24 show that some failure and recovery paths remain unsafe despite the restart passes.

Accessibility evidence

The source audit J25 found no JLabel.setLabelFor association in DesktopShellFrame. Accessible names exist for some sync/list controls, but the planner caption-to-field association still needs work. Keyboard use, screen-reader announcements and text scaling were not manually checked. No general accessibility-compliance claim is made.

Check	Requirement	Status
M01 Live Java server SQLite restart	FR5, QR1	BLOCKED HERE
M02 Keyboard-only complete workflow	AC1	NOT RUN
M03 Screen reader field names and status feedback	AC1	NOT RUN
M04 Small display and text scaling	AC1, QR2	NOT RUN
M05 Live interrupted network and retry	FR5, QR1, QR2	NOT RUN
M06 Unsynced work warning on exit	QR2	NOT RUN

M01 is blocked here because Maven and the SQLite JDBC dependency are unavailable. Other remaining checks require the actual final GUI and selected live client/server pair. Step-by-step scripts for these checks are in REMAINING_CHECKS.md. Existing PASS statements in server/README.md are treated as teammate claims rather than independent results from this run.

10. Run instructions traceability and handover

From the repository root run:

java scripts/RunRole7Tests.java

Use JDK 21 or later. On Windows the included Run_Role7_Tests.cmd runs the same command. Exit code 1 is expected for the recorded baseline because ten assertions fail. The custom runner is not automatically invoked by mvn test.

To preserve the original evidence and put your own run in a new folder:

java scripts/RunRole7Tests.java . target/role7-my-rerun

Requirement trace example

FR5 and QR1 require offline work to remain available and synchronise safely. J24 persists an interrupted SYNCING record, reloads LocalOutboxRepository and observes retry queue size zero. The result is FAIL and is logged as D06. The recommended change is to recover interrupted records to a retryable state. A developer must implement that change and rerun J24 before it can be marked resolved.

Recommended redesign work

First agree the final client/server pair and correct its payload and acknowledgement contract (J16, J18, S11). Next make plan/outbox persistence recoverable and resume interrupted sync attempts (J21, J24). Then complete required-element validation and explicit label associations (J23, S06, J25). These recommendations are supported by test evidence. No completed redesign or participant evaluation is claimed here.

Files for the team

Recipient	Material
Representative	Testing PDF, REPORT_INSERT.md, raw evidence and AI-use declaration.
Role 2	TRACEABILITY_MATRIX.md, using existing FR1-FR6, QR1, QR2, SUS1 and AC1.
Roles 4 to 6	DEFECT_LOG.md, Java tests and remaining recovery/accessibility checks.
Role 7 student	Individual evidence draft; complete identity, verification and real peer evidence.

Sources and disclosure

Assessment sources: original-brief.pdf, pages 2-4; A4_Tasks_By_Role.pdf, page 1. Repository baseline: docs/design/Requirments.md; technical design section 3; docs/traceability.md; actual production code and resources. Source hashes and timestamps accompany the execution evidence.